Cryptocurrency and Digital Assets: A Positive Tool for Economic Growth in Developing Countries

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ABSTRACT: Today's global economy finds itself impacted by a web of production and consumption that has led to increasing environmental concerns and political governance conflicts. As problems arise, humans look progressively towards technology to find solutions — one of these methods being cryptocurrency as a vehicle for investment and exchange, especially in developing countries plagued by hyperinflation. While citizens of developed nations benefit from stable financial infrastructure, the rest of the world's population does not share these privileges and therefore is excluded from the traditional banking system. Inflation ravages citizens of Zimbabwe, Venezuela, Myanmar, Argentina, and many others, who are forced to find an alternative to cash. Contrary to popular skepticism, Bitcoin is not merely an experimental tool to get rich, it serves in this case as a lifeboat to help preserve individuals' financial resources in the face of a rapidly depreciating currency. Theoretically, it allows anyone, anywhere (with an internet connection) to store their value regardless of income or socioeconomic status. However, volatility and regulation risks continue to rise as Bitcoin usage increases. I intend to explore whether or not cryptocurrency can propel economic growth in developing countries. Through analyzing the advantages and disadvantages of its usage and illustrating a working system in my case study of Venezuela, I dive deep into the alternative financial infrastructure that Bitcoin provides. Finally, I argue that cryptocurrencies are a positive tool for growth in developing countries, provided that increasing adoption in the future will increase the financial literacy necessary to access the digital resources.

1 Introduction

Modern-day economies find that financial markets are characterized primarily by a rapid flow of information. Participants interact with this information in a plethora of ways, defined by their preferences and investment horizons, which gives rise to a complex system. In this century, human efforts to understand and improve the system are pioneered by technological and digital solutions. In 2009, blockchain technology surfaced with the creation of Bitcoin, a digital currency experiment. The growth of blockchain and its applications, most notably cryptocurrency, over the past decade has yielded much hope that these disruptive technologies will rewire the world's financial markets and thus, the world in which we live. Unfortunately, there still is much skepticism and confusion surrounding the market of cryptocurrencies; many believe Bitcoin to be a fad that will fade away in due time.

Over the last few years, the cryptocurrency market has developed immensely, from being entirely peripheral in the global economy and its financial markets to capitalizing at the level of a medium-sized stock exchange (Wątorek 2021). One may ask: how has this come to be, and why now? When examining other technologies that have risen recently such as smartphones, the internet, and 5G, there is one major thing in common — these advancements all intend to make current processes more efficient and easier to access. They provide tools to better what might not be working anymore or to fix the flaws that are hindering a system. The same applies to Bitcoin, the first and most popular cryptocurrency.

Bitcoin first found a mention two months after the financial crisis in 2008, an event that sparked fears of central banks abusing power and the economic instability that may follow (Saleem 2018). The founder, an anonymous person or group of people operating under the pseudonym Satoshi Nakamoto, intended for Bitcoin to limit political power through decentralization. The digital currency is governed by smart contracts that execute decisions and are unchangeable by any central power.

The idea here is that the adoption of cryptocurrency and its market's emergence is propelled by the perceived failings of traditional financial systems. It is associated with low trust in banks among citizens of a region, and with country-level inflation crisis (Marthinsen 2020). However, due to the novelty of this asset and its high volatility, many doubt that it can be used as a positive and long-lasting tool, especially in regions that already suffer under fragmented governance systems. The debate on whether cryptocurrency can benefit such kinds of societies continues to dominate the current political and economic climate.

An opportunity arises now to analyze the cryptocurrency market's recent evolution and characteristics, which can help determine if the advantages of adopting a digital asset such as Bitcoin outweigh the disadvantages. The analysis also aims to clarify to what extent these characteristics have made the cryptocurrency market similar to traditional markets such as those of stocks, bonds, commodities, and foreign exchange. In the case that the markets are similar enough, perhaps the future may see that the digital asset market is here to stay, alongside or even in the place of traditional financial markets.

As the blockchain revolution continues to expand, the question lies with the macroeconomic impact that cryptocurrency may have. Is cryptocurrency itself a positive or negative tool in developing countries and emerging markets?

2 Advantages of Digital Assets

Decentralization

The blockchain technology behind digital assets is ruled by the notion of decentralization. This means that no central entity may wield control or decision-making power over the rest of the network. The process is governed simultaneously by no one, and by everyone. No one needs to know or trust anyone else — instead, each member has access to the same data as everyone else and if any individual attempts to corrupt the data, the rest of the network will reject the changes (Hirsh and Alman 2020). Decision-making is achieved in as equitable a manner as possible, through a majority and a quorum of votes. The ultimate goal is to reduce points of weakness in systems where too much reliance on specific actors can lead to conflicts and abuse of power.

Through decentralization, the blockchain system was designed by Satoshi Nakamoto to be secure against tyranny and other political power imbalances. It calls on the basic properties of human interaction and attempts to give each individual as much agency as they can have before their preferences become too strong and start to overtake another's. The concept is largely democratic in that the network is immutable — individuals do not have the power to change or alter the system without a majority consensus of the people. This kind of governance system is unique to all other current operating systems, which are mostly centralized. Thus, its unique factor gives it lots of opportunities to succeed, but also to fail.

The size of Bitcoin's network and its hash rate have protected it from monopolization by any one miner or mining group, simply because possessing a majority of the network's computational power is unlikely. The hardware and energy required for a potential perpetrator to execute this attack would not only require over 15 billion USD but would require a lot of luck on their side, owing to the randomness of the mining process (Bitpanda 2022). Hypothetically, if one entity were to succeed with a 51% attack, it could attempt to withhold its own mined blocks and create a competing chain that would overtake the original blockchain. It could also refuse to accept blocks mined by other network participants or blacklist certain Bitcoin addresses (Bitpanda 2022). Fortunately, the design is coded to prevent such attacks as these, which is why the system prevails and Bitcoin continues to dominate the cryptocurrency market.

Financial Inclusion and Equity

At its core, Bitcoin's rise has driven the growth of a vast digital ecosystem of entrepreneurship and exchange. When people look at the viability of digital assets, there prevails an ongoing debate between those who support its adoption and those who find more disadvantages in it. Some fear the movement of "dark" money from illicit transactions as a result of this technology and its provision of anonymity for every user. However, it is undeniable that Bitcoin has spearheaded a digital revolution with significant economic impact.

The main component of the "pro" argument for its widespread adoption lies in the potential it has for extending financial inclusion (Greenwald 2021). This can be defined as individuals and corporations having access to useful and affordable financial products and services that meet their needs. These may include transactions, payments, and savings, delivered responsibly and sustainably. Fundamentally, the design of digital asset platforms includes aspects of equity that are not present in the traditional banking system. Because of this, the adoption of digital assets could offer an abundance of advantages for an economy. Since economic empowerment often comes from the ability to build credit, increase buying power, maintain savings, and more, one can assume that greater financial inclusion would boost these abilities (Greenwald 2021).

When looking at the modern ecosystem of banking and value-holding, it is important for an economy to prioritize the balance of privacy and inclusion within the financial system. Digital assets seem to fulfill this goal with blockchain's guarantee of anonymity and more equity among participants. Although some stipulate that the emergence of digital currencies is simply an opportunity for governments to take back power from citizens, it is useful to think about a recent example of how current systems inadequately serve the people, thus reinforcing the need to modernize (Greenwald 2021).

In March 2020, the coronavirus pandemic hit the United States, placing the government in immediate spotlight — how will one of the world's leading hegemons respond? The answer was to provide emergency stimulus payments to repair the economic damage. Although these payments rolled out quickly, they lacked the necessary structure to reach the citizens that needed them most. As a consequence, over 6% of American adults, unbanked, did not receive a direct deposit, and likely had trouble receiving any written checks from the US Federal Reserve (Greenwald 2021). Consequent stimulus checks that were rolled out periodically over the next years ran into similar inefficiencies. While the Federal Reserve injecting trillions of dollars into the financial system has reinforced global confidence in the dollar, the US economic structure has revealed fundamental weaknesses in its attempt to implement a long-term stimulus plan (Greenwald 2020). These limitations, combined with ongoing global technological movements, call for transformative policymaking and a bold shift to a future dominated by a digital economy.

From the promises of greater financial inclusion and faster payments in times of crises, it can be determined that economies would have more tools and a greater capacity to build, or rather, rebuild the economy and empower individuals with digital financial systems. The understanding of increased equity and efficiency is critical to illustrating the future adoption of digital assets.

Currency

Cryptocurrencies are largely experimental, given their high volatility and unstable value compared to fiat money. However, their design fulfills a lot of money properties that fiat does not. The principal question here is: what is money? A good currency should be a store of value, a medium of exchange, and a unit of account. To qualify as these things, it needs to have certain properties. Below is a breakdown of these properties and the mark given to the USD, gold, and Bitcoin on a standard A to F grading scale.

Properties of Good Money	US Dollar	Gold	Bitcoin
Durable	B	A	A
Portable	B-	D	A
Fungible	A-	A	A-
Verifiable	D	C	A+
Divisible	B	B-	A
Scarce	D	B	A+
Widely Acceptable	A+	C	D+

Figure 1. Currency Properties (Galaxy Digital 2021).

Analyzing this chart, there are a couple of key observations. Firstly, a good currency should be durable. Fiat money is printed on paper and is physically vulnerable to tears and weathering. A simple tear down the middle of a dollar bill renders its value null. A digital currency does not have this weakness, and is much more durable — as long as it exists online, it remains there. Second, it is understood that Bitcoin is much more portable than the dollar bill due to its digital properties, and can lead to more widespread accessibility and use. As valuable as gold is, it receives a D in this category because it is finicky to transport.

Two other categories that Bitcoin outperforms the dollar bill in are verifiability and scarcity. A currency that is verifiable means that it has security features. While modern cash systems have been able to identify counterfeit bills well, ordinary individuals cannot tell a fake note from a real note. This leads to scams and overall less security when exchanging dollar bills on an interpersonal level. Bitcoin, on the other hand, is protected by digital transaction verification on its blockchain, using miners to sign hashes that validate transactions on the network (Latto and Regan 2021). Transactions are generally irreversible, leading to strong data security. Bitcoin is ultimately resilient due to its network's decentralization and the fact that it takes too much computer power to hack the system.

In terms of scarcity, the Bitcoin system stipulates that a maximum of 21 million units can be mined, distributed periodically over the course of a couple decades (Bitpanda 2022). This creates value for the Bitcoin due to the resource's scarcity, unlike fiat currency which in theory, does not have a printing limit. While this may be more difficult in practice, in which a government relies on the ability to enforce monetary policy and add money to the economy during recessions, scarcity and worth have a direct relationship that is generally true for anything in the world that has value.

The dollar bill surpasses all other forms of currency in one category, which is acceptability. Because it works everywhere, it is widely consistent and trustworthy in a way that most cryptocurrencies are currently not. Fiat currency also has a long history of use that strengthens its current hold on the global economy. However, paper money is created by the government and has value because the law says so. Unlike gold or rye stone, it does not hold any actual value — its value comes from a shared belief, a law that people rarely question. Although cryptocurrencies have begun to gain more traction globally in the past couple of years, their acceptability level is still relatively low. Given the novelty of decentralized technology and the recency in which blockchain has been implemented in practice, much is still unknown, and the world will have to wait and see how cryptocurrency's usage pans out.

3 Economic Impact of Cryptocurrencies

Economic Impact through Blockchain Usage

On the technological side, blockchain offers several significant advantages over traditional forms of finance including lower cost and increases in operational efficiencies of traded markets, property records, and transaction processes. This has led to the massive expansion of digital asset usage, and as of March 2022, there are over 10,000 cryptocurrencies in circulation globally (Howarth 2022). Just around a decade ago in April 2013, there were only 7 cryptocurrencies (Howarth 2022). See below a graph depicting the growth of traded cryptocurrencies over the past decade.

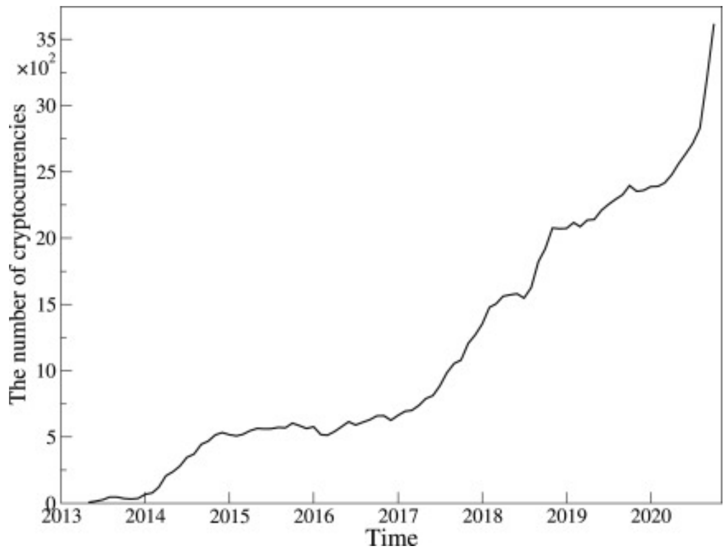

Figure 2. Number of Actively Traded Cryptocurrencies (Wątorek 2021).

So far, blockchain has improved financial institutions' cross-border transactions. The technology operates using encrypted distributed ledgers that provide trusted real-time verification of transactions without the need for intermediaries like banks. This makes it easier to reach international markets rather than strictly sticking to the national markets, allowing individuals to create relationships and foster trust with markets never before available. For developing nations, this helps expand their financial reach.

The technology itself can also be applied to a variety of other industries that can potentially unlock billions of dollars for those markets (Pelicoin 2020). Any sales platform can use blockchain to streamline transactions in a faster and more efficient way. Users thus feel confident knowing that their money is being transferred with a robust verification system. Additionally, government and public records can use blockchain to reduce paperwork and fraud while increasing accountability. Even individual companies can launch their own cryptocurrency or tokens to ensure services are being properly paid for by their consumer base. Blockchain can also be paired with other technologies such as cloud computing, to execute smart contracts and resist hacking.

Economic Impact on Job Markets

The upsurge in cryptocurrency usage has instigated the rise of an industry dedicated to supervising cryptocurrency exchanges globally. The number of jobs in the blockchain industry has increased from around 1,000 in 2016 to over 10,000 today (Consensys 2019). Engineering roles such as software engineering and full-stack development have been the most sought-after professionals in the industry. Coders are very valuable in the space, especially because fewer people have the skillset to learn nontraditional languages such as Solidity, which is the most used blockchain programming language. As cryptocurrency continues to be legalized outside of the western world, there is more expected job creation in the field, less unemployment, and thus further economic prosperity.

Economic Impact on Unstable Domestic Currencies

In countries where the domestic currency is constantly fluctuating, too volatile to be a stable store of value, living conditions plummet and financial management is difficult. These situations can be aided by the introduction of cryptocurrency. Inflation in particular, plagues the economies of Zimbabwe, Venezuela, Myanmar, and many others, who are forced to find an alternative to cash (Galaxy Digital 2021). For these citizens, cryptocurrency is not only the safer substitute, because it is not affected by their countries' governments' monetary policies, it can also act as a hedge against the risk they might take by holding cash. Additionally, corrupt governments cannot control the blockchain transaction system and thus provides individuals more independence to manage their financials without a central entity looking over their shoulder. In the face of a rapidly depreciating currency, something like Bitcoin can help preserve individuals' financial resources, regardless of their income or socio-economic status. This solution is not only effective and supportive of financial independence, it is also equitable. As long as the individual possesses a cell phone, they have the ability to transact cryptocurrency through mobile applications (Pelicoin 2020).

This topic will be expanded on in the section: Addressing Hyperinflation.

Economic Impact through Low Transaction Costs

Aside from Bitcoin — due to its proof-of-work consensus mechanism and high gas fees from miners — transaction costs for most cryptocurrency users are minimal to none. Because blockchain technology is decentralized and digital, not requiring investment into physical property, there are no extra costs that users are expected to pay (Pelicoin 2020). Unlike a branch of a bank, there is no need to pay costs like utility bills, rent, or employee wages. Overall, low transaction costs also encourage trust in the system of cryptocurrency, which sees more use in financial tools and a closer global economy (Pelicoin 2020).

Unfortunately, many popular cryptocurrencies have not yet tackled the problem of the variation of transaction fees imposed on users. The cost largely depends on supply and demand — for example, the Bitcoin block size is 1MB, meaning that miners can only confirm 1MB of transactions per block, with one block every ten minutes (Ebun-Amu 2021). As the blockchain gets overloaded, higher fees are paid. The cost also comes from moving assets from off-chain (cash) to on-chain (on the blockchain network) (Rapoza 2021). This year, the average daily transaction fee on the Bitcoin network has been as low as \$1.78 and as high as \$62, while for Ethereum they have been as low as \$1.59 and as high as \$70 (Vigna 2021). Besides these two largest cryptocurrencies, others see fees around a couple cents to a couple dollars.

To overcome these problems, many Layer-2 scaling solutions have been created, the most popular being Polygon, Optimistic Ethereum, and Arbitrum (Rapoza 2021). Users can use this bridge service to transfer their assets to a Layer-2 network in order to reduce gas fees and increase transaction speeds. Although this solution is slowly picking up speed, most retail investors do not trade in this world and look towards cryptocurrency exchanges instead. With exchanges, the only way to cut fees over time is to increase the number of users, which will come organically and gradually.

Economic Impact through Transparency

Transactions made through blockchain technology are automated and tracked on a digital ledger that cannot be manipulated or altered by people, companies, or governments. This ledger is public so that participants and observers alike can view the data, which establishes a high level of transparency. This allows for a level of trust between everybody involved, and the concept is called "trustlessness." Instead of trusting a third-party authority to oversee operations like a central bank, participants rely on an algorithm that publishes everything to the ledger. Behind this system is an algorithm that rewards a network of computers for validating transactions and drives them to remain honest.

Historically, the evolution of institutional exchange — from in-person bartering to digital finance — has called on trust as an important factor when supervising the use of money as a medium of exchange and a store of value. New decentralized trust protocols have introduced new developments in record-keeping in the practice of money exchange. The scenario in which code replaces law captures the benefits of formal social structures while simultaneously maintaining the freedom and scale of informal online groups. It promises participants both material and social values, the ability to exercise more autonomy for the individual, as well as a strong community of others that one can trust (Campbell-Verduyn 2018). Ultimately, this diminishes the risk of fraud and corruption (Pelicoin 2020).

This is particularly beneficial for developing countries and regions with corrupt governments that they do not trust. Citizens can thus rely on the utilitarian structure of cryptocurrencies to invest and transact with the global economy, which in turn boosts their economy and quality of life.

Economic Impact for Entrepreneurs

Due to its decentralized nature, cryptocurrency commands a global economy in which participants may exchange currency regardless of their citizenship. For entrepreneurs, this expands their audience from national to international, with whom funds can be exchanged without the hassle of exchange rates and international law. Furthermore, there are cryptocurrency companies that aim to assist business owners in Africa transact with European, American, and Asian companies, aiming to create financial coverage and liberation worldwide. The need to communicate across borders is now manifesting itself in financial means, and traditional financial institutions are not able to provide this as well as cryptocurrencies can. As more and more entrepreneurs turn towards digital solutions, they can engage in the investment, saving, and sending of money across borders, which may ultimately reframe global business practices. This will drastically alter the private market scene and internal capabilities of individual companies.

4 Addressing Hyperinflation

Although the traditional financial system has its flaws, one may ask why it must be reformed if it has been working decently well for centuries. Personal experience has lent the impression that at least in the United States, not many individuals run into problems with currency, savings, or banking. Understandably, citizens of developed nations benefit immensely from stable financial infrastructure. However, much of the world's population does not operate under the same structures and is therefore excluded from the benefits that citizens of developed nations enjoy (Galaxy Digital 2021). Poor economic governance, heavy money-printing, and large deficit spending have led many nations to be plagued by hyperinflation, which when sustained for extended periods, can cause entire currencies to collapse. As a hedge against massive inflation, cryptocurrencies come into play.

At the time of writing, hyperinflation wreaks devastating consequences in Zimbabwe, Sudan, and Venezuela, while its specter looms in yet a longer list of countries including Argentina, Ethiopia, and Turkey. A sharp decline in the value of the prevailing currency in most of these countries created immense social upheaval and pushed them to reconsider their economic foundations (Chohan 2021).

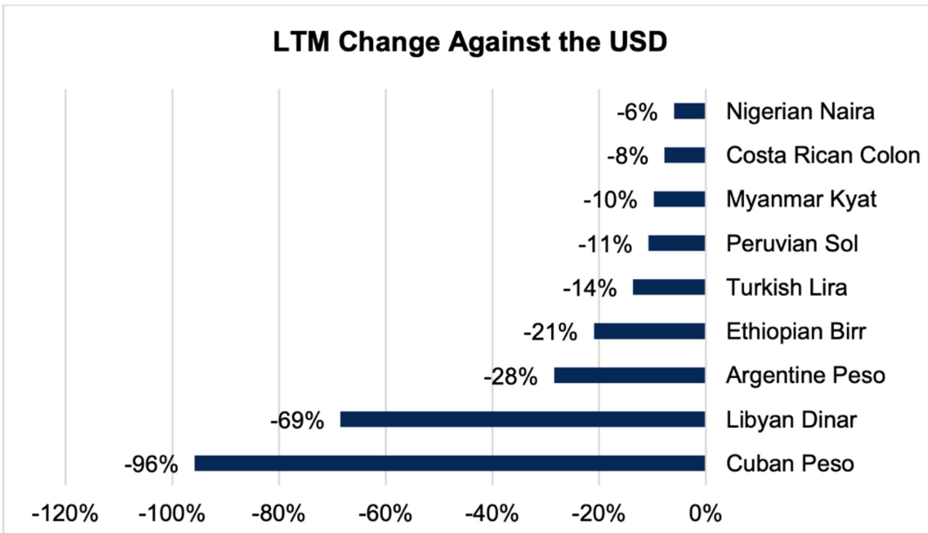

Figure 3. USD foreign exchange rate of select currencies over the twelve months ending May 5, 2021 (Bloomberg 2021).

Beyond wealth preservation, Bitcoin helps citizens of oppressive countries access money. The Human Rights Foundation points out around 4.2 billion people globally live under authoritarian regimes that use money as a tool for surveillance and state control. Their currency is often debased, cutting them off from the international financial system (Gladstein 2021). This is because hyperinflation is a form of disastrous economic shock that is unique to fiat money, created by central bank balance sheets (Clohan 2021). Fiat money has an intrinsic risk of oversupply, especially when the institutional independence of the central authority is compromised by political elements such as corruption. It works well when the society bears strong institutional mechanisms of independence, accountability, and oversight, which is true for most developed nations. Without these factors, it is hard to say.

Expansionary Monetary Policy

The current financial system that revolves around fiat currency allows central banks the power to adjust the base interest rate and print money to accomplish price stability. Printing money, however, has always been a cautious move for governments because if overdone, questions about the credibility of the currency and the central authority itself arise. There is a lot of financial freedom that comes with countries that are "currency-sovereign," meaning nations that can borrow significant amounts in international markets without fear of hyperinflation since they can simply pay back their debts by producing more of their own currency (Clohan 2021). Unfortunately, very few countries are truly currency-sovereign, and most risk this strategy backfiring on them in the FX markets from exorbitant money-printing.

The recent COVID-19 pandemic has seen many governments using monetary policy to print more money, expanding their balance sheets in response to the worldwide recession. For example, 40% of the US dollars that have ever existed came into being after the pandemic began in 2020 (Clohan 2021). Similar expansionary policies were put into effect in the wake of the crisis by the EU and other wealthy nations. This ability to print money with abandon is the luxury of very few currency-sovereign states and marks the ability of global hegemons to avoid hyperinflation. Remarkably, even attempts to stir inflation in the economy, such as those of Japan since 1989, find themselves failing (Clohan 2021). It seems that no amount of stimulus in the industrial world seems to generate inflation, even when deliberately done. Perhaps Marx's prediction of the falling rate of profit in late capitalism has come true.

Yet, the case for emerging markets is entirely different, where inflation poses a serious and persistent risk to the economy. No emerging market is sufficiently currency-sovereign, and many must borrow in large reserve currencies such as USD or EUR, impeding their ability to pay back debts (Clohan 2021). The ethos of their governments and central banks is often the focus of key investor sentiment, and the slightest doubts about their monetary stability and institutional integrity can magnify into full-fledged attacks on their currencies. In the event that emerging markets seek to print money to escape the shackles of debt, their currencies may quickly spiral out of control.

Role of Cryptocurrencies in Emerging Markets

How can cryptocurrencies seek to address hyperinflation in emerging market scenarios? Primarily, their decentralized nature protects them from being manipulated by any political powers or mismanaged by monetary authorities. No part of Bitcoin's system, for example, can be impacted by political or institutional tampering, which makes it an attractive store of value for emerging market citizens. Furthermore, Bitcoin's clearance network clears within one hour, while the traditional banking system requires days or even weeks (Clohan 2021). In terms of speed of transactions and access, a digital currency proves more efficient than paper money. The apolitical and predictable nature of its deflationary monetary policy means that a standard cryptocurrency can avoid hyperinflation entirely. For Bitcoin, the fixed amount of 21 million units ensures that it cannot be deployed for expansionary policies that may drastically grow the money supply. Rather, decision-making requires the majority consensus of a decentralized, distributed network.

However, not all cryptocurrencies are limited in their supply. Dogecoin is one example of a currency that is not fixed by any means. Most newer currencies have dictated that their supply will be contingent on the network's consensus. If the network decides further down the line that they would like to add a couple million units, it is up to the participants to vote on its execution. This is true for Bitcoin as well, though it is difficult to predict what will happen once all 21 million units are deployed. If a majority consensus exists among the node-holders of the Bitcoin network to extend the supply outwards, they could amend the protocols, and all relevant participants in the blockchain network must apply it. Alternatively, they could establish a "fork" and split the system into two: those following the old rules and those following the new ones.

While this aspect of the blockchain system is reasonably democratic and honors the preferences of as many participants as possible, it complicates things because in theory, the system could keep forking forever and end up with too many versions to manage. Fortunately, the majority consensus voting system is set up so that most preferences are captured and adequate discussion is had whenever a new change needs to be made (Campell-Verduyn 2018). The takeaway here is that one cannot say with certainty that the premise of a fixed supply will hold indefinitely for digital currencies.

In 2019, the New York Times published an op-ed written by Carlos Hernandez, a Venezuelan migrant to Colombia, titled “Bitcoin Has Saved My Family.” It evokes the lived experience of citizens in collapsing economies and dictatorships that find utility in cryptocurrency (Hernandez 2019). For Hernandez himself, the use of cryptocurrencies has not only helped him send money to Venezuela, it also played a large role in helping him carry wealth with him during his flight from the country. This resonates with other individuals seeking immediate relief from their home country’s state of affairs, who know they can successfully depend on the decentralized nature of Bitcoin to protect themselves. An important factor to note is that prerequisite conditions for these people would include sufficient computer literacy, as well as enough savings translatable into Bitcoin (Clohan 2021).

It is difficult, however, to assume that the usage of Bitcoin can benefit all users in developing nations — when the assumption comes with the idea that everyone has capital to begin with. Usually, it is the poor with meager savings that see them wiped out the quickest, particularly since they most likely do not have material assets to revert to in selling what they have: houses, cars, and especially the computers or smartphones that are required to access digital currency technology (Clohan 2021). Expecting people in this state of deprivation to hop onto a digital bandwagon is a bit improbable since they have less exchange value to convert into something like Bitcoin. The question for citizens of the developing world then becomes, is the opportunity cost of exchanging their native currency to a digital one worth it? The question for these developing economies as a whole is then, how can finding a safe haven in digital currencies be supported on a societal level? Currently, the switch to Bitcoin or other forms of cryptocurrency remains an individual decision and does not effectively show how nations can confront hyperinflation on a larger scale.

These are the considerations that must be made when analyzing the role of cryptocurrencies in the face of hyperinflation in developing countries. They are what economic policymakers, academics, and advocates must recognize when weighing their benefits against legitimate concerns that have been raised about cryptocurrencies (such as their legality, governance, and security) (Clohan 2021). For the time being, citizens in dire situations who are digitally literate and have a modicum of existing capital may see cryptocurrencies as a sound alternative to their depreciating local money. They will see advantages in inflation protection, conversion, digital storage, speed, and ease of access.

The Emerging Market in El Salvador

El Salvador is an example of an emerging market nation that has embraced cryptocurrency usage. On September 7, 2021 — with 2021 being the 20th anniversary of El Salvador’s dollarization of its economy — the country made Bitcoin legal tender (Casanova 2021). The government had launched a Bitcoin wallet called Chivo to facilitate cheap and efficient payments using Bitcoin’s Lightning Network, a protocol on the Bitcoin blockchain that enables fast transactions among participating nodes while reducing costs (Hussey 2022). Despite initial skepticism, Chivo's success over the past couple years has been rapid. In 2017, only 29% of the El Salvadoran population had bank accounts, but a mere two months after the new legal tender was introduced, 46% of the population — two million citizens — had downloaded Chivo (Thomas 2021). The convenience of the wallet allows Bitcoin to be used for micro-transactions such as buying a cup of coffee. It also supports free cross-border payments, which manifests in the form of helping migrant workers of El Salvador and their families to save more than $400 million in remittance fees per year (Casanova 2021). As previously discussed, good money requires being widely acceptable, and this is a factor that cryptocurrencies are currently lacking. In this specific case, declaring Bitcoin legal tender helps in expanding the currency's acceptability. All major chains and companies operating in El Salvador, such as McDonald's, must accept Bitcoin too (McEvoy 2021). The expansion of financial independence in El Salvador as a result of Chivo has influenced other Latin American countries to possibly follow in its footsteps.

Interestingly, the USD is accepted as legal tender in El Salvador as well (Casanova 2021). This illustrates the prevalent sentiment that many developing countries are looking for more stable, transparent, and reliable alternatives — it is not just countries with weak currencies. Avoiding the additional source of volatility that currency devaluations bring has urged emerging markets to actively seek alternatives.

5 Case Study: Venezuela

Venezuela is a prime example of a developing country, inundated with inflation, that has turned towards cryptocurrency as a solution. This case study aims to break down the unfolding of events that has led to Venezuelan citizens building a stronger and more stable financial ecosystem for themselves. The country's success with finding a working alternative in Bitcoin has influenced other nations suffering similar problems to follow suit.

The movement of Venezuelans turning to cryptocurrency was driven primarily by the launch of the petro by the Maduro regime. The petro was an attempt at a national currency by the presidency under Nicolás Maduro, created to evade US financial sanctions. In early 2018, the Trump administration was pressuring the Venezuelan government through sanctions, and Maduro's advisors retaliated by launching petro with a huge propaganda campaign and with some help from Russia (Rendon 2021). There was not much proof that petro existed as a usable technology, and citizens' distrust grew. The petro is subject to the regime's control, unable to be mined, and is unconstitutionally backed by a basket of natural resources including oil, iron, diamonds, and gold (Rendon 2021). There is limited public information about the technology behind petro or how it is used, even though it ironically claims to be an open and public cryptocurrency. Because of the clear lack of transparency in the petro system, it is anything but open and public. While the Maduro regime has been using petro to pay pensioners and retirees and has even made gas stations receive payments in petro, there is little evidence of transactions being made in petro (Rendon 2021). It is a financial instrument that disobeys many of the values that blockchain ethos supports such as decentralization and depoliticization.

Citizens of Venezuela, reasonably suspicious of petro, and living in this 2,600% inflation economy with authoritarian monetary policies, had to turn elsewhere for solutions (Rendon 2021). Cryptocurrencies built on blockchain technology that trade internationally offer a variety of solutions. These benefits include preserving the value of Venezuelans' money while protecting it from their government and offering ways to send humanitarian aid more efficiently and securely.

Open and public cryptocurrencies use cryptographic algorithms to record and secure information that is stored on a blockchain. Bitcoin is an example of an open and public cryptocurrency and is not issued, controlled, or maintained by any central authority. In countries like Venezuela, this is a pivotal element that shields the digital currency from manipulation, government interference, and false transactions (Rendon 2021). There are five main reasons why Venezuelans are adopting these, instead of the petro.

1. *Stability*.

In addition to a failing economy and hyperinflation, it is difficult to make day-to-day transactions. Cash is scarce and traditional payment networks take hours due to the network being overloaded. Although most do not attribute stability to cryptocurrency because volatility remains high, there are long-term benefits of holding bitcoin rather than bolivars or even US dollars. The latter two are difficult to transact due to domestic regulations and international sanctions. The bitcoin alternative does not have these obstacles and can be used immediately to preserve the value of savings or make payments.

2. *Remittances.*

Because more than 5 million people have fled the country in recent years, remittances have become a lifeline, reaching at least 35% of Venezuelan households (Rendon 2021). However, as oil revenues dry up and sanctions remain high, the cash-strapped regime has begun to regulate hard currency remittances from a growing Venezuelan diaspora. In response to this, decentralized, censorship-resistant cryptocurrencies could be used to mitigate the regime's controls while helping Venezuelans continue to send money home.

3. *Humanitarian Aid.*

The Maduro regime continues to reject international aid and Venezuela is increasingly cut off from the global financial system, which makes sending humanitarian aid a challenge. As a solution, some NGOs are exploring the use of cryptocurrency donations to buy and distribute food. For example, Bitcoin Venezuela feeds about 2,000 people daily through Bitcoin donations. Not only is this donation process quick, it incurs almost no cost from intermediaries, which is key in the humanitarian space due to scarce resources. Another example is Acció Solidaria, an NGO that combats HIV and other illnesses in the country, partnering with The Giving Block to create their own digital wallet to receive cryptocurrency donations (Rendon 2021).

4. *Security.*

Venezuela is one of the most unsafe countries in Latin America. The average citizen suffers from petty theft, sometimes even at the hands of the police, those meant to protect them. Refugees and migrants are especially vulnerable to having their money stolen while they travel throughout the region. Bitcoin and other cryptocurrencies guarantee safety when it comes to protecting citizens' money, as they are stored in a digital wallet that is password protected.

5. *Financial Inclusion and International Sanctions*.

Today, about 90% of remittance transactions go through international banking services such as Zelle (Rendon 2021). In spite of this, most Venezuelans do not have enough USD to open an American bank account to access services like Zelle. International bank accounts are generally difficult for Venezuelan citizens to use due to sanctions and overcompliance issues that have caused many to shut down. This is where open and public cryptocurrencies come in, offering a direct, cost-efficient, and more inclusive payment mechanism.

Ultimately, Bitcoin has begun the revolution of cryptocurrency use cases in developing countries, and many others will begin to explore the prospect. While this case study has investigated a working system in Venezuela, cryptocurrencies are by no means an immediate solution to all humanitarian and political challenges. Limited access to the internet, digital devices, lack of education and awareness on how to use the technology, and uncertainty of international regulations are some challenges to overcome.

6 Risks

The literature on digital assets is ever-expanding with ideas on where our global financial system is headed. Although nothing is known for certain, especially where technology is concerned, there are definite problem areas that need to be addressed in the next iteration of digital finance.

Although cryptocurrency and its underlying technology increase financial inclusion more than fiat currency does, it needs to expand financial literacy (Greenwald 2021). As previously mentioned, the prerequisites for using digital currencies are having internet access, a working digital device, and knowledge of how to use it. Even with the necessary infrastructure, consumers who do not understand how the system operates will find themselves less protected by the technology and more vulnerable to potential attacks on their assets. The development of education and infrastructure may come naturally once more people enter the ecosystem and start to use cryptocurrency, but as of now, there are limited actions to take by current users to help increase these forms of access in developing countries.

There is also ongoing skepticism surrounding cryptocurrency due to its facilitation of dark money flows and money laundering. While blockchain technology ensures privacy among its users and their transactions for many good reasons, it also enables bad actors to take advantage of and use cryptocurrency for illegal reasons. Because the system is intrinsically built to be unregulated, with no central authority overseeing anything, this is a difficult problem to address. A possible solution to this requires Know Your Customer and built-in regulatory guidelines to ensure beneficial ownership (Greenwald 2021). Know Your Customer is the mandatory process of identifying and verifying the client's identity when they open an account and while they are actively using the account. This can help decrease illegal activity in the system.

In terms of international regulation, a lot of it is still up in the air due to uncertainty of global preferences. To begin with, blockchain was built to resist regulation by central authorities, and working around that is something governments continue to struggle with. Some central governments opt for more surveillance-heavy banking and payment process, while others are holding their ground for privacy (Greenwald 2021). It is understood that values of a particular digital system need to be hashed out in detail before the technology becomes too widespread, or people will find themselves entrenched in a chaos-ridden ecosystem.

Another disadvantage of cryptocurrency in its current state is the carbon-intensive processes of its operation. Mining Bitcoin is a very energy-intensive process and it is estimated that one Bitcoin transaction takes 1,544 kWh to complete, or the equivalent of approximately 53 days of power for the average US household (Greenwald 2021). The carbon footprint of mining Bitcoin has become more significant over the years, especially since many of these transactions occur in places where coal power is still prominent. For example, the energy consumption of the Bitcoin blockchain in China, where around half of Bitcoin's miners are located, is expected to exceed the total annualized greenhouse gas emission output of the Czech Republic and Qatar by 2024 due to their coal-intensive electric grid (Jiang 2021). Fortunately, newer cryptocurrencies are addressing this issue through a shift from mining to staking, a technology that offers much less energy consumption (Greenwald 2021). Ethereum, the second-largest cryptocurrency after Bitcoin, has been working on its staking technology since 2021 and will upgrade to Ethereum 2.0 soon. The future of digital assets may see more staking, which does not require much energy input to begin with, as well as greater adoption of sustainable, less carbon-intensive energy sources to power cryptocurrencies.

7 Conclusion

Ultimately, the birth and growth of digital assets have made it possible for nations suffering from hyperinflation to find another solution by adopting cryptocurrency. To succeed, cryptocurrencies must be able to provide nations with a stable store of value and a foundation for long-term growth and financial development. As more nations adopt it, citizens' trust in the currency and its overall usage will increase. Additionally, cryptocurrency allows these citizens to trade freely across borders with citizens of more well-off countries, creating a level of economic equality (Pelicoin 2022).

After analyzing the large-scale impacts that cryptocurrencies have on the global economy as well as identifying its benefits to emerging markets that suffer from hyperinflation — specifically Venezuela, its overall advantages can be seen in a tangible application. In theory, the decentralized nature of blockchain technology makes accessing the currency much more equitable and efficient, but citizens of developing countries often run into financial literacy obstacles. The infrastructure and education required to expand financial literacy in these countries become an effort that moves in tandem with increased use. Given the novelty of this space, that is a development that remains to be seen, perhaps in the near future. Fundamentally, cryptocurrencies are a positive tool for growth in developing countries, provided that higher adoption in the future will increase financial literacy and thus, access to the necessary digital resources.

Despite the risks identified in the previous section, each of the challenges shows that digital assets have a great role to play in the next evolution of the global economy. Whether it is cryptocurrencies, stablecoins, or CBDCs (central bank digital currency), each has properties that will increase financial inclusion. As authoritarian governments continue to operate around intense citizen surveillance in payment systems, democratic governments must continue to champion the empowerment of the individuals in their economies and allow citizens in developing nations to access the same financial resources. This can lead to long-term economic growth and faster recovery in emergencies such as the recent COVID-19 pandemic.

Undoubtedly, the underlying technology of this ecosystem has the capacity to revolutionize the way financial and commodity markets operate. Even in terms of energy, many fail to remember that the operation of the current banking system and gold mining accounts for a far greater amount of energy use than Bitcoin mining (Greenwald 2021). This means that a partial replacement of the current banking system with digital assets may end up creating a discount on energy expenditure for financial systems, all while increasing speed and efficiency. While it is yet unknown whether cryptocurrencies can or will sufficiently replace the current fiat money ecosystem, digital assets and their economic benefits will certainly continue to expand alongside the traditional system.

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